

MVP2 UI/UX Design System Specification

MVP2 UI/UX Design System Specification

Helix VPN Cross-Platform Design System

Version: 1.0.0

Date: July 2025

Status: Draft for Implementation

Scope: Desktop (macOS, Windows, Linux), Mobile (Android, iOS, HarmonyOS), Web, Browser Extension

Frameworks: Tauri v2 (Desktop), Flutter (Mobile), React (Web/Extension)

Table of Contents

1. [Design System Overview](#)
 2. [Color System](#)
 3. [Typography](#)
 4. [Layout Grid](#)
 5. [Component Library](#)
 6. [Screen Specifications](#)
 7. [Interaction Patterns](#)
 8. [Iconography](#)
 9. [Animation & Motion](#)
 10. [Platform Adaptations](#)
-

1. Design System Overview

1.1 Design Philosophy

The Helix VPN design system follows a **clean, minimal, security-focused** philosophy. The UI prioritizes clarity and trust, reflecting the app's core purpose: protecting user privacy. Every design decision serves the goal of making security feel effortless and approachable.

Core Principles:

Principle	Description	Implementation
Clarity	Every element has a clear purpose	Minimal chrome, focused layouts, no decorative noise
Trust	Security should feel tangible	Color-coded states, clear visual feedback, transparent status
Efficiency	Primary actions are immediately accessible	One-tap connect, gesture shortcuts, contextual options
Consistency	Unified experience across all platforms	Shared design tokens, consistent interaction patterns
Accessibility	Usable by everyone	WCAG 2.1 AA compliance, screen reader support, keyboard navigation

1.2 Design Tokens Approach

Design tokens are the single source of truth for all visual properties. They enable cross-platform consistency while allowing platform-specific adaptations.

```
tokens/
├─ colors/                # Color palette tokens
│   ├── primary.json      # Brand colors (teal/cyan)
│   ├── semantic.json     # Status/state colors
│   ├── surfaces.json     # Background/surface colors
│   └─ text.json          # Text colors
├─ typography/           # Type scale tokens
│   ├── scale.json        # Size ramp (Display -> Overline)
│   └─ family.json        # Font family per platform
├─ spacing/              # Spacing scale tokens
│   └─ scale.json         # 4px base grid
├─ elevation/            # Shadow/depth tokens
│   └─ shadows.json
└─ motion/               # Animation tokens
```

```

├── duration.json
├── easing.json
└── radii/                # Border radius tokens
    └── scale.json

```

Token Format (JSON/CSS Variable):

```

/* Example: Primary token cascade */
--helix-primary-50: #E0F2F1; /* Teal tint */
--helix-primary-100: #B2DFDB;
--helix-primary-200: #80CBC4;
--helix-primary-300: #4DB6AC;
--helix-primary-400: #26A69A;
--helix-primary-500: #00897B; /* Primary brand */
--helix-primary-600: #00796B;
--helix-primary-700: #00695C;
--helix-primary-800: #004D40;
--helix-primary-900: #00332C;

```

1.3 Cross-Platform Consistency Strategy

The design system employs a **"unified core with native skin"** approach:

- **Shared Design Tokens:** Colors, spacing, typography scales are consistent across all platforms
- **Platform-Native Components:** Widgets adapt to each platform's design language (Material 3, Cupertino, Fluent, Silica)
- **Unified Interaction Patterns:** Connection flows, state transitions, and gestures are consistent
- **Platform-Specific Chrome:** Navigation, menus, and system integration follow platform conventions

DESIGN SYSTEM				
Core Layer (Shared)				
- Color tokens, spacing, iconography				
- Connection state semantics				
- Information architecture				
- Accessibility standards				
Desktop	Mobile	Web	Browser	
(Tauri)	(Flutter)	(React)	Extension	
- Web UI	- Material3	- Tailwind	- Popup UI	
- System tray	- Cupertino	+ shadcn	- Options	
- Menu	- ArkUI		- Content script	
- Menu	- Silica			

1.4 Accessibility First (WCAG 2.1 AA)

All components and screens must meet **WCAG 2.1 Level AA** compliance:

Requirements:

- Color contrast ratio $\geq 4.5:1$ for body text
- Color contrast ratio $\geq 3:1$ for large text (18pt+) and UI components
- All interactive elements have visible focus indicators (focus-visible)
- All form inputs have associated `<label>` elements
- Dynamic content changes announced via `aria-live` regions
- Full keyboard navigation support (Tab order, Enter/Space activation)
- Touch targets minimum 44x44dp/dpx (mobile)
- `prefers-reduced-motion` respected for all animations
- Screen reader tested with NVDA (Windows), JAWS (Windows), VoiceOver (macOS/iOS), TalkBack (Android)

2. Color System

2.1 Primary Palette (Teal/Cyan)

The primary palette uses teal/cyan tones to evoke trust, security, and technological sophistication.

Token	Hex	HSL	Usage
--helix-primary-50	#E0F2F1	174° 52% 91%	Lightest tint, backgrounds
--helix-primary-100	#B2DFDB	174° 47% 78%	Hover states, light fills
--helix-primary-200	#80CBC4	174° 42% 65%	Secondary highlights
--helix-primary-300	#4DB6AC	174° 38% 51%	Accent elements
--helix-primary-400	#26A69A	174° 62% 40%	Links, active states
--helix-primary-500	#00897B	174° 100% 27%	Primary brand color
--helix-primary-600	#00796B	174° 100% 24%	Primary hover
--helix-primary-700	#00695C	174° 100% 21%	Primary pressed
	#004D40		Strong emphasis

Token	Hex	HSL	Usage
--helix-primary-800	#00332C	174° 100% 15%	Deepest shade
--helix-primary-900		174° 100% 10%	

Secondary Cyan Accent:

Token	Hex	Usage
--helix-accent-400	#00BCD4	Secondary accent, data viz
--helix-accent-500	#00ACC1	Accent hover
--helix-accent-600	#0097A7	Accent pressed

2.2 Semantic Colors

Semantic colors convey meaning universally across all platforms.

State	Token	Hex	RGBA	Usage
Connected	--helix-connected	#4CAF50	rgba(76,175,80,1)	VPN connected, success states
Connecting	--helix-connecting	#FF9800	rgba(255,152,0,1)	In-progress, warning states
Disconnected	--helix-disconnected	#F44336	rgba(244,67,54,1)	VPN disconnected, offline
Error	--helix-error	#D32F2F	rgba(211,47,47,1)	Critical errors, destructive
Warning	--helix-warning	#F57C00	rgba(245,124,0,1)	Caution, attention needed
Info	--helix-info	#1976D2	rgba(25,118,210,1)	Informational messages

Semantic Opacity Variants:

```
--helix-connected-bg: rgba(76, 175, 80, 0.12);
--helix-connecting-bg: rgba(255, 152, 0, 0.12);
--helix-disconnected-bg: rgba(244, 67, 54, 0.12);
--helix-error-bg: rgba(211, 47, 47, 0.12);
--helix-warning-bg: rgba(245, 124, 0, 0.12);
--helix-info-bg: rgba(25, 118, 210, 0.12);
```

2.3 Latency Color Coding

Latency indicators use a traffic-light system:

Latency Range	Color	Hex	Semantic
< 50ms	Green	#4CAF50	Excellent
50-100ms	Amber	#FF9800	Good
100-200ms	Orange	#F57C00	Fair
> 200ms	Red	#F44336	Poor

2.4 Dark Theme Colors

```
.dark {  
  /* Surfaces */  
  --helix-bg-primary: #0A1929;           /* Deepest background */  
  --helix-bg-secondary: #132F4C;        /* Cards, panels */  
  --helix-bg-tertiary: #1E4976;         /* Elevated surfaces */  
  --helix-bg-elevated: #243B55;         /* Modals, dialogs */  
  
  /* Text */  
  --helix-text-primary: #F0F4F8;        /* Primary text */  
  --helix-text-secondary: #8BA3B8;      /* Secondary/muted */  
  --helix-text-tertiary: #5A7A94;       /* Placeholder, disabled */  
  --helix-text-disabled: #3D5A80;       /* Disabled text */  
  
  /* Borders */  
  --helix-border-default: #1E4976;  
  --helix-border-subtle: #132F4C;  
  --helix-border-focus: #00897B;  
}
```

2.5 Light Theme Colors

```
.light {  
  /* Surfaces */  
  --helix-bg-primary: #F0F4F8;          /* Background */  
  --helix-bg-secondary: #FFFFFF;        /* Cards, panels */  
  --helix-bg-tertiary: #E8EDF2;        /* Elevated surfaces */  
  --helix-bg-elevated: #FFFFFF;        /* Modals, dialogs */  
  
  /* Text */  
  --helix-text-primary: #0A1929;        /* Primary text */  
  --helix-text-secondary: #4A6375;     /* Secondary/muted */  
  --helix-text-tertiary: #8BA3B8;      /* Placeholder */  
  --helix-text-disabled: #B0C4D4;      /* Disabled text */  
  
  /* Borders */  
  --helix-border-default: #C8D8E4;  
  --helix-border-subtle: #E2EBF2;
```

```
--helix-border-focus: #00897B;
}
```

2.6 Surface & Background Colors Summary

Surface	Dark Theme	Light Theme
App background	#0A1929	#F0F4F8
Card/panel	#132F4C	#FFFFFF
Elevated (modal)	#1E4976	#E8EDF2
Input field	#0D2137	#FFFFFF
Selected item	rgba(0,137,123,0.15)	rgba(0,137,123,0.10)
Hover overlay	rgba(255,255,255,0.05)	rgba(0,0,0,0.04)
Pressed overlay	rgba(255,255,255,0.08)	rgba(0,0,0,0.08)
Divider	#1E4976	#E2EBF2
Scrim (modal backdrop)	rgba(0,0,0,0.7)	rgba(0,0,0,0.5)

3. Typography

3.1 Font Family Selection

Platform	Font Stack	Fallback
Desktop (Tauri)	system-ui, -apple-system, BlinkMacSystemFont, 'Segoe UI', Roboto, sans-serif	sans-serif
Android (Flutter)	Roboto (Material 3 default)	Device sans-serif
iOS (Flutter)	SF Pro Text/Display (Cupertino)	Device sans-serif
HarmonyOS	HarmonyOS Sans	Device default
Aurora OS	Sailfish Silica	System font
Web/ Extension	Inter, system-ui, -apple-system, sans-serif	sans-serif

Monospace (Technical Data): | Platform | Font | Usage |
|---|---|---| | All | JetBrains Mono, 'Fira Code', 'SF Mono', monospace |
IP addresses, connection stats, protocol info, keys |

3.2 Type Scale

The type scale follows a 1.25x ratio (major third) with a 14px base.

Token	Size	Weight	Line Height	Letter Spacing	Usage
Display	36px / 2.25rem	300 (Light)	1.2	-0.5px	Connection status numbers
Headline	28px / 1.75rem	600 (SemiBold)	1.3	-0.25px	Screen titles (desktop)
Title Large	22px / 1.375rem	600	1.3	0	Dialog titles, section headers
Title Medium	18px / 1.125rem	600	1.4	0.15px	Card titles, sub-screens
Title Small	16px / 1rem	500 (Medium)	1.4	0.1px	List section headers
Body Large	16px / 1rem	400 (Regular)	1.5	0.5px	Primary body text
Body Medium	14px / 0.875rem	400	1.5	0.25px	Default body, descriptions
Body Small	12px / 0.75rem	400	1.5	0.4px	Captions, metadata
Caption	11px / 0.6875rem	500	1.4	0.5px	Labels, timestamps
Overline	10px / 0.625rem	600	1.4	1.5px	Category labels, all-caps

3.3 Font Weights

Weight	Name	Usage
300	Light	Large display numbers, hero text
400	Regular	Body text, descriptions
500	Medium	Buttons, list items, emphasis
600	SemiBold	Headlines, titles, section headers
700	Bold	Critical labels, connection status

3.4 Monospace Type Scale (Technical Data)

Token	Size	Weight	Usage
Mono Large	18px	400	Connection time, large stats
Mono Medium	14px	400	IP addresses, server info
Mono Small	12px	500	Protocol badges, key fragments
Mono Caption	11px	400	Debug info, logs

4. Layout Grid

4.1 Desktop Layout Grid

The desktop UI uses a responsive grid system with a minimum width of 360px and maximum of 1440px.

Property	Value
Min width	360px (small popups)
Max width	1440px (full admin panel)
Main window	420px (connection) / 520px (servers) / 640px (settings)
Gutter	24px
Column count	12 (fluid)
Margin	24px (desktop), 16px (compact)

Desktop Window Sizes: | Screen | Width | Height | Purpose |
|---|---|---|---| | Main (connected) | 420px | 600px | Primary interface
| | Main (disconnected) | 420px | 480px | Compact state | | Server
Selection | 520px | 700px | Full server list | | Settings | 640px |
700px | Tabbed settings | | Connection Details | 420px | 550px |
Stats overlay |

4.2 Mobile Layout Grid

Platform	System	Base Unit
Android	Material 3	8dp grid
iOS	Cupertino	8pt grid
HarmonyOS	ArkUI	4vp grid

Mobile Safe Areas:

- Top: Respect SafeArea top inset (notch, status bar)

- Bottom: Respect SafeArea bottom inset (home indicator, nav bar)
- Horizontal: 16dp/dpx padding (mobile), 24dp (tablet)

4.3 Spacing Scale (4px Base)

All spacing uses a 4px base unit for consistency.

Token	Value	Pixels	Usage
--space-0-5	0.125rem	2px	Icon gaps, hairline spacing
--space-1	0.25rem	4px	Tight component internal padding
--space-2	0.5rem	8px	Default component internal padding
--space-3	0.75rem	12px	Small component margin
--space-4	1rem	16px	Default padding, card gutters
--space-5	1.25rem	20px	Medium spacing
--space-6	1.5rem	24px	Section padding, dialog margins
--space-8	2rem	32px	Large section gaps
--space-10	2.5rem	40px	Section separators
--space-12	3rem	48px	Major section padding
--space-16	4rem	64px	Hero spacing, connection button margin
--space-20	5rem	80px	Large visual separation

4.4 Responsive Breakpoints

Token	Width	Target
xs	< 380px	Extension popup (minimum)
sm	380-640px	Extension popup, mobile PWA
md	640-768px	Tablet PWA, small panels
lg	768-1024px	Tablet landscape, compact desktop
xl	1024-1280px	Desktop standard
2xl	1280px+	Desktop wide, admin panel

4.5 Border Radius Scale

Token	Value	Usage
--radius-none	0	Sharp edges (tables, dividers)
--radius-sm	4px	Small tags, badges

Token	Value	Usage
--radius-md	8px	Buttons, inputs, small cards
--radius-lg	12px	Cards, panels
--radius-xl	16px	Large cards, bottom sheets
--radius-2xl	24px	Modals, dialogs
--radius-full	9999px	Pills, avatars, connection button

5. Component Library

5.1 Desktop Components (Tauri/WebView)

5.1.1 Button Variants

Variant	Background	Text	Border	Hover
Primary	#00897B	#FFFFFF	none	#0070C0
Secondary	transparent	#00897B	1px solid #00897B	rgba(0,136,203,0.5)
Ghost	transparent	#F0F4F8	none	rgba(0,136,203,0.1)
Danger	#D32F2F	#FFFFFF	none	#B71C1C
Disabled	rgba(255,255,255,0.12)	rgba(255,255,255,0.38)	none	-

Button Specs:

- Min height: 40px (standard), 32px (compact)
- Horizontal padding: 24px (standard), 16px (compact)
- Border radius: 8px (standard), 4px (compact)
- Font: Body Medium, 500 weight
- Icon + text gap: 8px

5.1.2 Connection Toggle (Desktop)

[Power Icon]
CONNECTED
US East (23ms)

- Size: 140px x 140px circular
- Connected: Fill #00897B with #4CAF50 glow pulse
- Connecting: Animated spinner on #FF9800 ring
- Disconnected: #F44336 with subtle shadow
- Border: 4px solid ring (color = state)
- Glow: 20px blur, state color at 30% opacity, pulsing

- Icon: 48px, white
- Label below: Body Large, 600 weight, state color

5.1.3 Server List Item

```
+-----+
| [FLAG]  US East - New York      [Fav] WG |
|          23ms  * * * *  34% load          |
+-----+
```

- Height: 64px
- Left: Flag icon (36x36px, 8px radius) + 12px gap
- Title: Body Large, 500 weight
- Subtitle: Body Small, secondary color
- Right: Favorite star + Protocol badge + Chevron
- Divider: 1px bottom border
- Selected state: Primary tint background (10% opacity)
- Hover: Hover overlay

5.1.4 Protocol Badge

```
+-----+
|  WG  |
+-----+
```

- Size: auto-width x 20px
- Background: Primary at 10% opacity
- Text: Primary color, Caption weight 500
- Border radius: 4px
- Padding: 0 6px
- Variants: "WG" (WireGuard), "OV" (OpenVPN), "IK" (IKEv2)

5.1.5 Latency Indicator

Color	Size	Animation
Green (<50ms)	8px circle	Static
Amber (50-100ms)	8px circle	Subtle pulse
Orange (100-200ms)	8px circle	Pulse
Red (>200ms)	8px circle	Fast pulse

- Displayed as colored dot + "23ms" label
- Dot has 4px colored glow (same color, 40% opacity)

5.1.6 Settings Card/Section

- Background: bg-secondary
- Border radius: 12px
- Padding: 16px
- Section header: Title Small, primary color, uppercase

- Item height: 56px
- Divider between items: 1px border-subtle
- Left icon: 24px, primary or secondary color

5.1.7 Input Fields

+-----+		
	Label	
	+-----+	
	Placeholder text	[]
	+-----+	
	Helper text	
+-----+		

- Height: 48px
- Background: bg-primary (dark) / #FFFFFF (light)
- Border: 1px border-default
- Border radius: 8px
- Padding: 0 16px
- Font: Body Medium
- Focus: 2px border-focus outline
- Error: 2px #D32F2F outline + red helper text
- Disabled: 38% opacity

5.1.8 Modal/Dialog

- Min width: 360px, Max: 560px
- Background: bg-elevated
- Border radius: 16px
- Shadow: 0 24px 48px rgba(0,0,0,0.4)
- Header: Title Large, 600 weight
- Content: Body Medium
- Actions: Right-aligned, Primary + Secondary buttons
- Backdrop: Scrim color with fade-in 150ms

5.1.9 Toast Notification

+-----+		
	[Icon] Message	[X]
+-----+		

- Width: auto (max 400px)
- Height: 48px
- Background: bg-elevated
- Border left: 4px (semantic color)
- Border radius: 8px
- Shadow: 0 8px 24px rgba(0,0,0,0.3)
- Duration: 4000ms (auto-dismiss)
- Animation: Slide in from top-right, 300ms ease-out

5.1.10 System Tray Menu (Desktop)

+-----+		
	HelixVPN - Connected	
	US East (23ms)	

	[ICON] Connect	
	[ICON] Disconnect	

	[ICON] Select Server...	
	[ICON] Preferences...	

	[ICON] Quit	
+-----+		

- Follows platform menu conventions
 - macOS: NSStatusItem with template icon
 - Windows: NotifyIcon with context menu
 - Linux: AppIndicator/StatusNotifierItem
 - Items: 28px height, 16px icon, 12px gap
 - Separator: 1px line
-

5.2 Mobile Components (Flutter)

5.2.1 Connection Button (Large Circular)

- Size: 140dp diameter
- Shape: Circle with radial gradient glow
- Connected: #00897B fill, #4CAF50 pulsing outer ring (20dp spread)
- Connecting: #FF9800 fill, rotating spinner overlay
- Disconnected: #F44336 fill, static
- Icon: 48dp, white, centered
- Shadow: 0 20px 40px state-color at 30%
- Animation: Pulse scale 1.0 -> 1.08, 1500ms ease-in-out (connected only)

5.2.2 Server Selection Tile

- Height: 72dp (Material 3 ListTile)
- Leading: Country flag circle (40dp)
- Title: Body Large, 500 weight
- Subtitle: Body Small (latency dot + "23ms" + load %)
- Trailing: Protocol chips + favorite icon
- Selected: Primary tinted background
- Ripple: Platform default

5.2.3 Bottom Sheet

- Initial height: 50% screen
- Max height: 90% screen
- Background: Surface color
- Top handle: 4dp x 32dp, #8BA3B8, 2dp radius
- Border radius: 24dp top corners
- Entry animation: Slide up 300ms, ease-out
- Drag to dismiss: Velocity threshold 300dp/s

5.2.4 Tab Bar

- Height: 48dp (Android) / Cupertino navigation bar (iOS)
- Background: Surface color
- Indicator: 2dp height, Primary color
- Active text: Primary, 500 weight
- Inactive text: Secondary color, 400 weight
- iOS: Segmented control or bottom Cupertino tab bar

5.2.5 Switch/Toggle (Platform-Adaptive)

- Android: Material 3 Switch with Primary track
- iOS: CupertinoSwitch with Primary active color
- Track: 32dp x 52dp (iOS) / 28dp x 48dp (Android)
- Thumb: 24dp (iOS) / 20dp (Android)
- Active: Primary color track
- Inactive: Surface variant track

5.2.6 Text Field (Adaptive)

- Android: Material 3 OutlinedTextField with Primary focus
- iOS: CupertinoTextField with subtle border
- Height: 56dp (Android) / 44dp (iOS)
- Border radius: 8dp
- Focus: 2dp Primary outline
- Error: 2px #D32F2F outline

5.2.7 Card

- Background: Surface color
- Border radius: 16dp
- Elevation: 0 (flat), shadow for elevated
- Padding: 16dp
- Content: Title + body + optional actions

5.2.8 Speed Graph/Sparkline

- Height: 80dp
- Width: Parent full width
- Download line: #00897B, 2dp stroke

- Upload line: #00BCD4, 2dp stroke, dashed
- Fill: Gradient fade to transparent (line color 20%)
- Grid: Dotted horizontal lines at 25/50/75%
- Labels: Mono Caption, secondary color
- Update interval: 1000ms

5.2.9 Data Usage Indicator

+-----+		
Download	Upload	
[==== 1.2 GB]	[==== 340 MB]	
45% of limit		
+-----+		

- Two-column layout
- Value: Mono Medium, 500 weight
- Bar: 4dp height, full border radius
- Fill: Primary (download), Accent (upload)
- Background: Surface variant

5.2.10 Quick Settings Tile (Android)

- Size: 1x1 tile (standard QS tile)
- Active: Primary color background, white icon
- Inactive: Surface color, secondary icon
- Label: "Helix VPN" / "Helix VPN: ON"
- Toggle animation: Color crossfade 200ms
- Long-press: Opens app

5.3 Aurora OS Components (Silica)

5.3.1 Pulley Menu

- Trigger: Pull down from top edge
- Background: Primary color, dark theme
- Items: 56px height, white text
- Icon: 24px left-aligned
- Selection highlight: rgba(255,255,255,0.15)
- Animation: Slide down from top, items stagger 50ms

5.3.2 Context Menu

- Trigger: Long-press on list items
- Background: Elevated surface
- Items: 48px height, Body Medium
- Divider: 1px subtle
- Animation: Fade in 150ms + scale from anchor point

5.3.3 List Item (Silica Style)

- Height: 80px (standard), 64px (compact)
- Background: Transparent
- Primary label: Body Large, text-primary
- Secondary label: Body Small, text-secondary
- Icon: 32px, left-aligned, 16px padding
- Divider: 1px bottom border-subtle
- Pressed: rgba(255,255,255,0.08) overlay

5.3.4 Text Field (Silica Style)

- Background: Surface color
- Border: 1px border-default, no radius (Silica rectangular)
- Height: 48px
- Font: Body Medium
- Placeholder: text-tertiary
- Focus: 2px Primary underline
- Error: 2px #D32F2F underline

5.3.5 Slider

- Track height: 4px
- Active track: Primary color
- Inactive track: Surface variant
- Thumb: 20px circle, Primary fill, 2px white border
- Value label: Floating tooltip above thumb

5.3.6 Dialog (Silica)

- Width: 90% screen, max 480px
- Background: Elevated surface
- Border radius: 8px (Silica style)
- Header: Title Medium, 600 weight
- Content: Body Medium
- Actions: Full-width buttons stacked
- Entry: Fade 200ms + slight scale

5.3.7 Cover Page

- Size: 1:1 square (app cover)
 - Background: Gradient from Primary-800 to Primary-900
 - Connection status: Centered large dot (12px) + state text
 - Server: Caption text below status
 - Animation: Dot pulse glow matching state color
 - Ambiance: Respects system ambiance settings
-

6. Screen Specifications

6.1 Desktop Screens

6.1.1 Main Window (Connected State)

```
+-----+
| [TRAY] Helix VPN [ ] [X] |
+-----+
|
| [Server Selector Bar]
| [US Flag] US East - New York >
|
|           +-----+
|           | [POWER] | <- Large
|           | CONNECTED| toggle
|           +-----+
|           (glowing ring)
|
| Connected
| 23ms latency
|
| IP: 203.0.113.45
| Protocol: WireGuard
| Duration: 02:34:18
|
| [Speed graph sparkline]
|
| Download: 1.2 GB Upload: 340MB
|
+-----+
```

Specs:

- Window: 420px x 600px
- Background: bg-primary
- Server selector: 64px height, bg-secondary, 12px radius
- Connection button: 140px circle, centered
- Status text: Title Medium, #4CAF50
- Info panel: bg-secondary card, 12px radius, Mono Small text
- Speed graph: 80px height, full width minus 32px padding
- Data usage: Two-column flex layout

6.1.2 Main Window (Disconnected State)

- Window: 420px x 480px (compact)
- Connection button: 140px, #F44336 fill
- Status: "Disconnected", Title Medium, #F44336
- Info panel hidden

- Prompt: "Tap to connect securely", Body Small, secondary

6.1.3 Server Selection

- Window: 520px x 700px
- Search bar: Sticky top, 48px height
- Quick connect: "Optimal Location" tile, 56px
- Section headers: "Favorites", "All Locations" - Overline style
- Server list: Scrollable, items 64px height
- Country groups: Collapsible accordion
- Latency sort: Default (ascending)
- Selected: Primary tint background + checkmark

6.1.4 Settings (Tabbed)

- Window: 640px x 700px
- Tabs: General | Connection | Account | Advanced
- Tab height: 40px
- Content: Settings cards in vertical scroll
- Each card: bg-secondary, 12px radius, 16px padding
- Card title: Title Small, primary color
- Card items: 56px height rows

6.1.5 Connection Details / Stats

- Overlay modal or separate tab
- Real-time: Download/upload speed (Display size)
- Connection time: Mono Large, counting up
- Protocol info: Badge + details
- Server info: Flag + name + IP
- Encryption: Cipher info in monospace
- Graphs: 5-minute rolling sparkline

6.2 Mobile Screens

6.2.1 Connection (Home Screen)

```
+-----+
| [Menu] Helix VPN  [Cog]  |
+-----+
|
|  [Server Bar]
|  [US] US East          23ms |
|
|          +-----+
|          |  0  |
|          +-----+
|        CONNECTED
|
```

[Connection Stats Card]
IP: 203.0.113.45
Time: 02:34:18
Protocol: WireGuard
[Speed Sparkline]
DL: 45 Mbps UL: 12 Mbps

+-----+

Specs:

- Full screen, SafeArea respected
- App bar: 56dp, transparent or surface
- Server bar: 72dp, tappable, navigates to server list
- Connection button: 140dp circle, centered vertically
- Status text: Title Medium below button
- Stats card: bg-secondary, 16dp radius, 16dp margin
- Bottom section: Speed + data, Body Small

6.2.2 Server Selection

- Full screen with search AppBar
- Search field: 56dp, bg-secondary
- Quick connect: "Optimal Location" tile, 72dp
- Favorites section: Horizontal scroll chips
- Country groups: Expandable
- Items: 72dp ServerListTile
- Sort options: Latency (default), Load, Name, Distance
- Filter: Protocol, Streaming, P2P

6.2.3 Settings

- Sections: Connection, Auto-Connect, Security, App, Account, About
- Section header: Overline, Primary color
- Items: Material ListTile, 56dp height
- Toggle items: Switch trailing
- Navigation items: Chevron trailing
- Sub-screens: Push navigation (slide in from right)

6.2.4 Split Tunneling Configuration

- Tab bar: Off | Include | Exclude
- Mode explanation: Info banner at top
- App list: Checkbox tiles with app icons
- Search: Filter installed apps
- Save button: Bottom sticky, full width

- iOS limitation: Show warning about app enumeration restrictions

6.2.5 Connection Stats

- Real-time speed: Display size, Mono
- Session duration: Mono Large, counting
- Total data: Download/Upload counters
- Graph: 5-minute sparkline, full width
- Protocol details: Expandable card
- Server info: Location, IP, load

6.2.6 Support / Help

- Search FAQ at top
 - Categories: Cards in grid (2 columns)
 - Contact options: Chat, Email
 - Diagnostics: "Send logs" button
 - Version info: Caption at bottom
-

7. Interaction Patterns

7.1 Connection Flow Animation

DISCONNECTED	->	CONNECTING	->	CONNECTED
Red dot		Spinning ring		Green glow
Static		Amber color		Pulse animation
				Checkmark appears

Phase 1: Initiate (0-500ms)

- Button press: Scale to 0.95, 100ms
- Ring color transition: Red -> Amber, 200ms
- Spinner appears: Fade in + start rotation

Phase 2: Handshake (500ms-2000ms)

- Spinner: Continuous rotation (360deg/1500ms)
- Status text: "Connecting..." fade in
- Optional: Progress ring around button

Phase 3: Success (2000ms+)

- Ring color: Amber -> Green, 300ms
- Spinner fades out, checkmark scales in (0->1, 400ms bounce)
- Glow effect: 20px blur, Green at 30%, pulse loop begins
- Status: "Connected" text + server info slide up

- Stats panel: Slide in from bottom, 400ms

Disconnect Flow:

- Reverse of connect
- Green -> Red transition, 300ms
- Stats panel slides out
- Glow fades

7.2 State Transitions

Transition	Duration	Easing	Visual
Button press	100ms	ease-out	Scale 0.95
Color change	200ms	ease-in-out	Crossfade
Panel slide	400ms	cubic-bezier(0.4, 0, 0.2, 1)	translateY
Fade in	300ms	ease-out	opacity 0->1
Scale in	400ms	spring(1, 80, 10)	scale 0->1
Page push (mobile)	300ms	ease-in-out	translateX
Bottom sheet	300ms	ease-out	translateY

7.3 Pull-to-Refresh

- Trigger: Pull down past 64dp on scrollable lists
- Indicator: Circular progress, Primary color
- Release threshold: 80dp
- Animation: Rotation + arc sweep
- Haptic: Light impact at threshold (iOS)
- Completion: Snap back 200ms

7.4 Swipe Actions (Mobile)

Direction	Action	Target
Swipe left (server)	Delete favorite	Server list item
Swipe right (server)	Quick connect	Server list item
Swipe left (setting)	Reset to default	Settings item

- Action width: 72dp
- Background: Semantic color (green for connect, red for delete)
- Icon: 24dp, white
- Full swipe: Auto-execute action

7.5 Long-Press Menus

- Trigger: 500ms long press
- Menu: Context menu / Bottom sheet (adaptive)
- Items: Copy IP, View details, Quick actions
- Haptic: Medium impact on trigger
- Dismiss: Tap outside or back gesture

7.6 Keyboard Navigation (Desktop)

Key	Action
Tab	Move focus forward
Shift+Tab	Move focus backward
Enter/Space	Activate focused element
Escape	Close modal/dialog/menu
Ctrl+K	Quick connect/disconnect
Ctrl+Shift+S	Open server selection
Ctrl+,	Open preferences
Ctrl+Q	Quit application

- Focus ring: 2px Primary outline, 2px offset
- Focus visible: Only on keyboard navigation (not mouse click)

7.7 Screen Reader Support

- All interactive elements have descriptive aria-label
- Connection state changes announced via aria-live="polite"
- Server selection announces: "Connected to US East, 23 milliseconds latency"
- Status updates: "Connection lost, reconnecting"
- Landmark regions: main, navigation, complementary
- Skip links for main content

8. Iconography

8.1 Icon Set Requirements

Unified icon set across all platforms:

- Style: Outlined (line weight 1.5px-2px), rounded caps
- Size variants: 16px, 20px, 24px, 32px, 48px
- Color: Inherits text color by default
- Platform icon libraries:
 - Desktop/Web: Lucide React (or Phosphor Icons)
 - Mobile (Android): Material Symbols Outlined

- Mobile (iOS): SF Symbols
- Aurora: Sailfish Silica icons

8.2 Core Icon Inventory

Icon	Name	Usage	Platforms
Power	power / power_settings_new	Connection toggle	All
Shield	shield / shield.checkered	Security status	All
Shield Check	shield-check	Protected state	All
Globe	globe	Server/ location	All
Server	server	Server list	All
Settings	settings / gearshape	Preferences	All
Chevron Right	chevron-right	Navigation	All
Chevron Down	chevron-down	Expand	All
Search	search / magnifyingglass	Search	All
Close	x / xmark	Close/dismiss	All
Check	check / checkmark	Selected	All
Star	star / star.fill	Favorite	All
Clock	clock	Duration/time	All
Speed	gauge / speedometer	Speed test	All
Download	download / arrow.down	Download data	All
Upload	upload / arrow.up	Upload data	All
Lock	lock	Secure/ encrypted	All
Lock Open	lock-open	Unsecured	All
Wifi	wifi	Network	All
Wifi Off	wifi-off	No network	All
Alert	alert-triangle	Warning	All
Info	info	Information	All
Trash	trash	Delete	All
Refresh	refresh-cw	Refresh/retry	All
Menu	menu	Hamburger menu	All
Help	help-circle	Support	All

Icon	Name	Usage	Platforms
Log Out	log-out	Sign out	All

8.3 Platform-Specific Icons

System Tray Icon (Desktop)

State	macOS	Windows	Linux
Disconnected	Template PNG (B&W)	.ico 16x16	PNG 22x22
Connected	Template + green dot	.ico + overlay	PNG + badge
Connecting	Template + amber dot	.ico + overlay	PNG + badge
Error	Template + red dot	.ico + overlay	PNG + badge

- macOS: Use template image for dark mode compatibility
- Badge dot: 8px diameter, bottom-right offset
- Tooltip: "HelixVPN - Connected (US East)"

App Icon (All Platforms)

- Primary: Teal (#00897B) shield/globe motif
- Format: SVG source, exported to all platform sizes
- Sizes: 16, 32, 48, 72, 96, 128, 144, 192, 256, 512, 1024px
- Shape: Adaptive (Android), Rounded corners (iOS), Square (desktop)
- Background: Transparent or Primary-900

Notification Icon

- Android: Monochrome vector drawable, white on notification background
- iOS: App icon with tinted overlay
- Desktop: Platform-native notification with app icon

Quick Settings Icon (Android)

- Size: 24dp (standard QS icon)
- Style: Monochrome, white fill
- States: Disconnected (outline), Connected (filled)
- Label: "Helix VPN"

Extension Icon (Browser)

- Toolbar: 16px, 32px (PNG)

- Extension page: 48px, 128px
 - Manifest: 16, 32, 48, 128px
 - Color: Follows connection state badge overlay
-

9. Animation & Motion

9.1 Connection Animation

Button Pulse (Connected State):

Animation: scale 1.0 -> 1.08 -> 1.0
Duration: 1500ms
Easing: ease-in-out
Iteration: Infinite
Target: Outer glow ring + shadow

Ripple Effect (On Connect):

Origin: Button center
Animation: Scale from 0 to 4x, opacity 0.5 -> 0
Duration: 600ms
Easing: ease-out
Color: Primary at 20%

Status Glow:

Connected: Green glow, 20px blur, pulse opacity 0.2 -> 0.4
Connecting: Amber glow, rotating sweep gradient
Disconnected: No glow, static red
Error: Red pulse, 800ms (faster than connected)

9.2 Page Transitions

Desktop:

- Modal: Fade in backdrop (200ms) + scale content (300ms, spring)
- Page switch: Crossfade 200ms
- Drawer: Slide from right, 300ms ease-out

Mobile (Android):

- Push: Slide in from right (enter) + fade out (exit), 300ms
- Pop: Slide out to right (exit) + fade in (enter), 300ms
- Bottom sheet: Slide up 300ms, drag dismiss with velocity

Mobile (iOS):

- Push: Slide in from right, 350ms, Cupertino transition
- Pop: Slide out to right, 350ms

- Modal: Cover vertical, 400ms

9.3 Loading States

Skeleton Loading:

- Background: Surface color
- Shimmer: Linear gradient sweep, 1200ms
- Shape: Rounded rectangles matching content
- Trigger: Data fetch > 300ms

Spinner:

- Size: 24px (small), 40px (medium), 56px (large)
- Stroke: 3px, Primary color
- Duration: 1000ms per rotation
- Track: Surface variant, 20% opacity

Progress Bar:

- Height: 4px (linear), 48px (circular)
- Fill: Primary gradient
- Track: Surface variant
- Animation: Smooth width transition, 300ms

9.4 Micro-Interactions

Interaction	Trigger	Animation	Duration
Button hover	Mouse over	Scale 1.02, subtle lift	150ms
Button press	Mouse down	Scale 0.98	100ms
Toggle switch	Tap	Thumb translateX + color	200ms
Checkbox	Tap	Checkmark stroke draw	200ms
Radio	Tap	Dot scale in (spring)	200ms
Card hover	Mouse over	translateY -2px, shadow increase	200ms
List item tap	Touch	Ripple from touch point	400ms
Refresh	Pull	Rotation + arc sweep	1000ms
Toast	Trigger	Slide in from top-right	300ms
Copy feedback	Tap copy	Checkmark flash	500ms

9.5 Reduced Motion Support

All animations respect `prefers-reduced-motion: reduce`:

```
@media (prefers-reduced-motion: reduce) {  
  *, *::    animation-duration: 0.01ms !important;  
    animation-iteration-count: 1 !important;  
    transition-duration: 0.01ms !important;  
  }  
}
```

Reduced Motion Fallbacks:

- Pulse -> Static glow (no animation)
 - Page transitions -> Instant
 - Spinners -> Static indicator + text "Loading..."
 - Ripples -> Instant opacity change
 - Graphs -> Static (no draw animation)
-

10. Platform Adaptations

10.1 macOS

Aesthetic Principles:

- Safari-style minimal chrome
- Translucent sidebar support (vibrancy)
- Rounded window corners (system default)
- System font: SF Pro

Native Integration:

- MenuBarExtra: Icon + status tooltip, click to show window
- Native menu bar: HelixVPN > About, Preferences, Services, Quit
- Window: Title bar with traffic lights, no custom chrome
- Keyboard shortcuts: Cmd+K (connect), Cmd+, (prefs), Cmd+Q (quit)
- Notifications: NSUserNotification with action buttons
- Auto-start: Login item via SMLoginItemSetEnabled

Design Adjustments:

- Increased border radius (16px cards vs 12px default)
- Subtle shadows (macOS design language)
- Sidebar navigation for settings (3-column layout optional)
- Translucent toolbar: `window.vibrancy = 'under-window'`

10.2 Windows

Aesthetic Principles:

- Fluent Design influence (Mica/Acrylic materials where supported)
- Segoe UI font family
- Corner radius: 8px (Windows 11) / 0px (Windows 10)
- System context menu styling

Native Integration:

- System tray: NotifyIcon with context menu
- Jump list: Connect, Disconnect, Recent servers
- Notifications: Windows Toast with inline actions
- Installer: MSI with WiX, UAC elevation
- Auto-start: Registry Run key

Design Adjustments:

- Snap layouts: Window supports Windows 11 snap zones
- Title bar: Custom with Mica material on Win11
- Settings: NavigationView-style left sidebar
- Density: Slightly more compact (8px spacing vs 12px)

10.3 Linux

Aesthetic Principles:

- Adaptable to GTK/Qt themes
- Respects system accent color where possible
- Follows freedesktop.org HIG
- Font: System default (usually Cantarell, Noto Sans)

Native Integration:

- System tray: AppIndicator3 / StatusNotifierItem
- Desktop file: Categories=Network;Security;VPN;
- Notifications: dbus notification server
- Auto-start: .desktop file in ~/.config/autostart
- Theming: Detect GTK theme, adapt colors

Design Adjustments:

- Respect GTK_THEME for color adaptation
- Flat design (no shadows) for GNOME compatibility
- Header bar: Integrate with CSD if available
- Density: Configurable (compact/comfortable)

10.4 Android (Material You)

Aesthetic Principles:

- Material 3 design system
- Dynamic color: Primary derived from wallpaper (Android 12+)
- Rounded shapes: 12dp cards, 24dp dialogs
- Elevation: Surface tint instead of shadows

Native Integration:

- Quick Settings tile: 1x1 toggle in notification shade
- Foreground service: Persistent notification with controls
- App shortcuts: Long-press launcher icon for connect/disconnect
- Widget: 4x1 home screen widget with status + toggle
- Biometric: Fingerprint/face unlock for app access
- Always-on VPN: `SUPPORTS_ALWAYS_ON` metadata

Design Adjustments:

- Dynamic color: Use `dynamicColorScheme()` when available
- Fallback: Static teal palette on Android <12
- Navigation: Bottom nav bar (3 items: Connect, Servers, Settings)
- Status bar: Transparent with dark/light icon adaptation
- Edge-to-edge: Extend content behind system bars

10.5 iOS (Cupertino Design)

Aesthetic Principles:

- Cupertino design language
- SF Symbols iconography
- Blur effects: System materials (`systemThinMaterial`, etc.)
- Rounded corners: System default radii
- Font: SF Pro Text/Display

Native Integration:

- Control Center: VPN toggle in Settings > VPN
- Widget: Small + medium home screen widgets
- Shortcuts: Siri Shortcuts for "Connect VPN"
- Biometric: Face ID / Touch ID gate
- Push: APNs for service alerts
- URL scheme: `helixvpn://connect`, `helixvpn://disconnect`

Design Adjustments:

- Navigation: `CupertinoNavigationBar` with blur
- Lists: `CupertinoFormSection.insetGrouped` for settings
- Buttons: `CupertinoButton` with system styling

- Switches: CupertinoSwitch (slimmer than Material)
- Pickers: CupertinoPicker / modal bottom picker
- Modals: CupertinoModalPopup with sheet presentation

10.6 HarmonyOS (ArkUI Design Language)

Aesthetic Principles:

- ArkUI design language
- Smooth rounded corners: 16vp cards, 24vp dialogs
- Subtle gradients and depth
- Font: HarmonyOS Sans

Native Integration:

- Service widget: Form card for home screen quick connect
- VpnExtensionAbility: System VPN integration
- Notification: HMS Push Kit for alerts
- Distributed capability: Cross-device sync (future)
- Biometric: System fingerprint/face

Design Adjustments:

- Colors: Teal primary with HarmonyOS accent adaptation
- Cards: Elevated with subtle shadow
- Typography: HarmonyOS Sans throughout
- Animations: Smooth spring physics (ArkUI default)
- Dialogs: Centered with blur backdrop

10.7 Aurora OS (Silica)

Aesthetic Principles:

- Silica UI design language
- Dark-first design (ambiance integration)
- Edge gestures: Swipe from edges for navigation
- Pulley menus: Pull down for context actions
- Transparency and blur effects
- Font: Sailfish Silica

Native Integration:

- Cover page: 1x1 app cover with connection status
- Pulley menu: Pull down for connect/disconnect/server
- Ambiance: Adapt UI tint to system ambiance
- Sailfish notifications: Platform-native alerts
- D-Bus: System integration for network state

Design Adjustments:

- Transparency: Silica glass effect for panels
- Navigation: No bottom nav, use page stack + gestures

- Input: Silica-styled text fields with underline focus
 - Lists: Sailfish list items with press overlay
 - Primary action: Often in pulley menu, not always visible
 - Page transitions: Silica horizontal slide with depth
-

Appendix A: Token Reference Quick Sheet

Colors

```
/* Primary */
#00897B (500), #00796B (600), #00695C (700)
/* Accent */
#00BCD4 (400), #00ACC1 (500)
/* Semantic */
#4CAF50 (connected), #FF9800 (connecting), #F44336 (disconnected)
#D32F2F (error), #F57C00 (warning), #1976D2 (info)
/* Dark theme bg */
#0A1929 (primary), #132F4C (secondary), #1E4976 (tertiary)
/* Light theme bg */
#F0F4F8 (primary), #FFFFFF (secondary), #E8EDF2 (tertiary)
```

Spacing

4px, 8px, 12px, 16px, 20px, 24px, 32px, 40px, 48px, 64px

Border Radius

4px (sm), 8px (md), 12px (lg), 16px (xl), 24px (2xl), 9999px (full)

Animation

Fast: 150ms, Base: 200ms, Slow: 300ms, Page: 400ms
Easing: cubic-bezier(0.4, 0, 0.2, 1)
Spring: type-spring, damping: 20, stiffness: 300

Appendix B: Implementation Checklist

Phase 1: Foundation

☐

Define all design tokens in JSON

- ☐ Implement color system (dark + light themes)
- ☐ Set up typography scale per platform
- ☐ Configure spacing scale
- ☐ Implement border radius tokens

Phase 2: Core Components

- ☐ Button variants (primary, secondary, ghost, danger)
- ☐ Connection toggle with animation
- ☐ Server list item
- ☐ Protocol badge
- ☐ Latency indicator
- ☐ Input fields
- ☐ Modal/dialog
- ☐ Toast notification

Phase 3: Screen Layouts

- ☐ Desktop main window (connected/disconnected)
- ☐ Desktop server selection
- ☐ Desktop settings (tabbed)
- ☐ Mobile connection screen
- ☐ Mobile server selection
- ☐ Mobile settings
- ☐ Mobile split tunneling

Phase 4: Interactions

- ☐ Connection flow animation
- ☐ State transitions
- ☐ Pull-to-refresh
- ☐ Swipe actions
- ☐ Long-press menus
- ☐ Keyboard navigation
- ☐ Screen reader support

Phase 5: Polish

- ☐ Icon set implementation
- ☐ Animation system
- ☐ Reduced motion support
- ☐ Platform-specific adaptations
- ☐ Accessibility audit
- ☐ Cross-platform consistency review

Document generated: July 2025

Version: 1.0.0-MVP2

Classification: Design System Specification

Total Platforms: 7 (macOS, Windows, Linux, Android, iOS, HarmonyOS, Aurora OS)